

PhD Research Proposal

Title: Interpretable Spatio-Temporal Machine Learning Models for Urban Air Quality and Health Impact Analysis

1. Introduction and Motivation

Air pollution is a major environmental and public health concern, particularly in urban areas where population density and emission sources are high. Exposure to pollutants such as PM_{2.5} and NO₂ is strongly associated with respiratory and cardiovascular diseases. Despite the availability of environmental monitoring systems, predicting air quality at fine spatial and temporal scales remains a significant challenge.

Recent advances in machine learning offer powerful tools for modeling complex environmental systems. However, many existing approaches are purely data-driven, lack interpretability, and do not explicitly connect environmental predictions with health outcomes. This limits their usefulness in real-world decision-making and policy design.

This research aims to develop **interpretable and robust machine learning models** that capture the spatio-temporal dynamics of urban air pollution while integrating insights into public health impacts.

2. Research Objectives

The main objective of this PhD is to design machine learning models that are both **predictive and interpretable**, with applications in environmental monitoring and health analysis.

Specific objectives include:

- Develop spatio-temporal models for air quality prediction using heterogeneous data
- Capture spatial dependencies between urban regions using structured representations
- Integrate environmental predictions with health-related indicators
- Improve interpretability of machine learning models for decision-making
- Explore scalable approaches for real-world deployment

3. Methodology

3.1 Data and Problem Formulation

Air quality prediction is formulated as a spatio-temporal learning problem where pollution levels evolve over time and across locations.

$$y_t = f(X_t, S; \theta) + \epsilon_t$$

Where:

- y_t : predicted air quality at time t
- X_t : input features (weather, sensors, etc.)

- S : spatial structure
- θ : model parameters
- ϵ_t : noise term

The study will combine:

- Environmental sensor data
- Meteorological data
- Urban spatial information
- (Where available) health-related datasets

3.2 Model Development

The research will explore models that capture both **temporal dynamics** and **spatial relationships**:

- Temporal models (e.g., LSTM, sequence-based learning)
- Spatial models using graph representations of urban areas
- Hybrid spatio-temporal architectures

Particular emphasis will be placed on **model efficiency and generalization**, especially in data-scarce environments.

3.3 Interpretability and Explainability

To ensure practical relevance, interpretability will be a central component:

- Feature importance analysis
- Explainable AI methods (e.g., SHAP)
- Visualization of spatial and temporal patterns

This will allow identification of key drivers of pollution and support informed decision-making.

3.4 Health Impact Analysis

A secondary component of the research is to explore the relationship between air quality and health outcomes:

$$H_t = g(y_t, Z_t)$$

Where:

- H_t : *health outcome at time t*
- y_t : air pollution level
- Z_t : additional factors (e.g., demographics, environment)

The goal is to:

- Identify correlations between pollution exposure and health risks
- Provide insights for public health interventions

4. Expected Contributions

This research is expected to contribute:

- Improved machine learning models for air quality prediction
- Methods for integrating environmental and health data
- Interpretable AI tools for environmental monitoring
- Insights relevant to sustainable urban development

5. Relevance to the Host Institution

This project aligns closely with research in **multidisciplinary AI**, particularly at the intersection of machine learning, sustainability, and public health. The focus on real-world impact and interpretable models is consistent with ongoing research activities within the department.

6. Work Plan

Year 1: Literature review, data collection, baseline models

Year 2: Development of spatio-temporal models

Year 3: Health analysis and interpretability

Year 4: Validation, publications, and thesis writing

7. Conclusion

This project proposes a focused approach to modeling urban air quality using machine learning, with an emphasis on interpretability and societal impact. By linking environmental prediction with health analysis, the research aims to contribute to more informed and sustainable urban decision-making.

8. References

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